

# QANDEEL

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Herd, don't harvest: live-release management of Blue Blubber jellyfish swarms after ENEC's early warning

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# We keep swarms out of the intake and return them to the sea alive

## Problem

Blue Blubber swarms (30–45 cm, 1–3 kg) drift onto seawater intakes. Clearing them takes heavy manual work and almost none survive.

## Solution

After ENEC’s alert, software plans the response; a bubble curtain at the breakwater gap holds the swarm out; a boom with a closed-bottom retention bag gathers it in the water; it is released alive down-current.

## Evidence

Working simulation on real 2025 Copernicus currents and ERA5 wind; dashboard; 16 automated tests; open code.

## Next

A 1:20 tank test (AED 300 kit) to measure what the model assumes, then a sea pilot with ENEC.

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swarms that truly arrived were covered, in a hindcast of 100 alerts on real 2025 Gulf data

77%

fewer curtain running hours than switching on for every alert

84–100%

of jellyfish kept by the boom with our retention bag (model), vs 0–54% for an open oil-spill boom

46%

of the energy of a fixed airflow, by setting air hour by hour from real currents and wind

# A large, local, hard-to-handle jellyfish

ENEC already detects swarms offshore. The open question is what to do next: manage high volumes with less manual work and less harm.

## The target: Blue Blubber

- One of the two dominant outbreak species on the UAE coast; reproduces year-round (Aravind 2024)
- 30–45 cm bell, ~1–3 kg each (Yang et al. 2018 mass formula)
- 87–100% sit in the top 1 m by day (Pitt & Kingsford)
- Large adults swim >10 cm/s: not just passive drifters

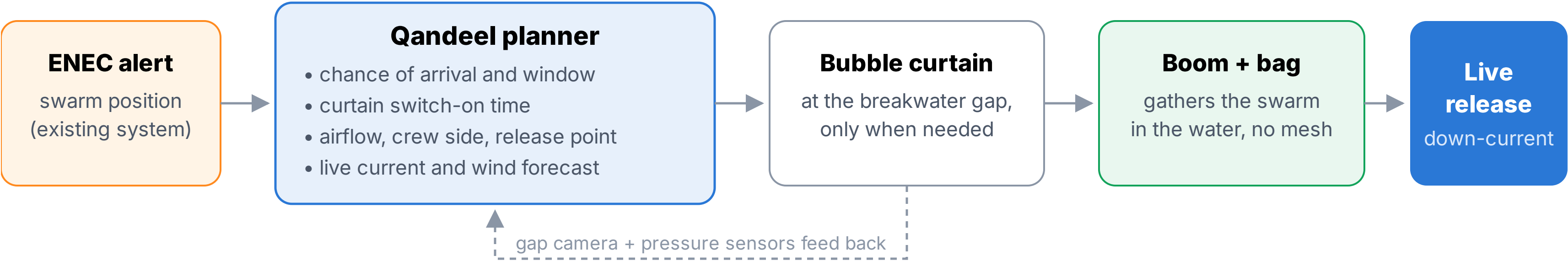
## What happens at intakes

- Swarms follow the intake flow through the breakwater opening and pile on screens
- Worldwide this has forced power reductions and shutdowns (e.g. Torness 2011, Oskarshamn 2013)
- In the region, ~300 t of jellyfish clogged intakes in Oman

## Why it is hard

- Manual raking and trucking, sometimes for days
- Jellyfish crushed on screens or discarded: almost none survive
- Cutting robots (e.g. JEROS) shred them into fragments
- No management method has ever been tested on this species

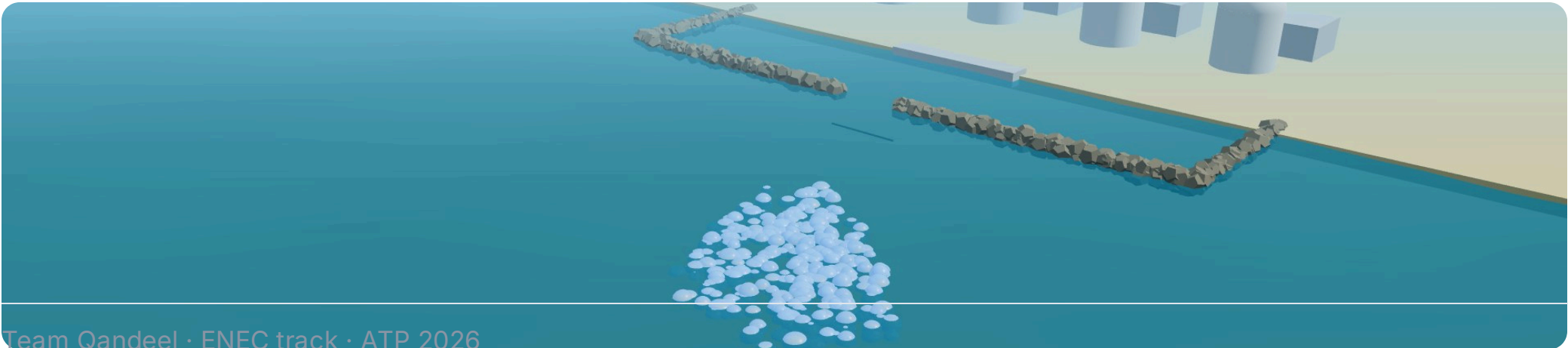
# Guard the one gap, move the water, keep every jellyfish in the sea



**1 • The gap is a chokepoint**  
Seawater, and the swarm, can only reach the intake through one opening.

**2 • Move the water, not the animal**  
A rising-bubble current holds them out; nothing is cut, pumped or netted.

**3 • Borrow proven oil-spill tools**  
Booms and boats exist today; we found and fixed why they leak jellyfish.



The breakwater leaves one opening to the intake. A swarm offshore must pass through it, so that is the one place we act. Image: our 3D model (stylised; not to scale).

# One swarm, from alert to live release

- 1

Alert.

ENEC reports a swarm, e.g. 15 km NNW. The planner receives it automatically.
- 2

Plan.

400 simulated drifts of 60 swimming jellyfish each, on the live current and wind forecast: 54% chance of arrival, window 10–77 h.
- 3

Prepare.

Curtain on at +7 h; crew staged on the approach side; release point chosen.
- 4

Deflect.

Curtain across the gap holds the swarm outside; airflow follows the current.
- 5

Gather.

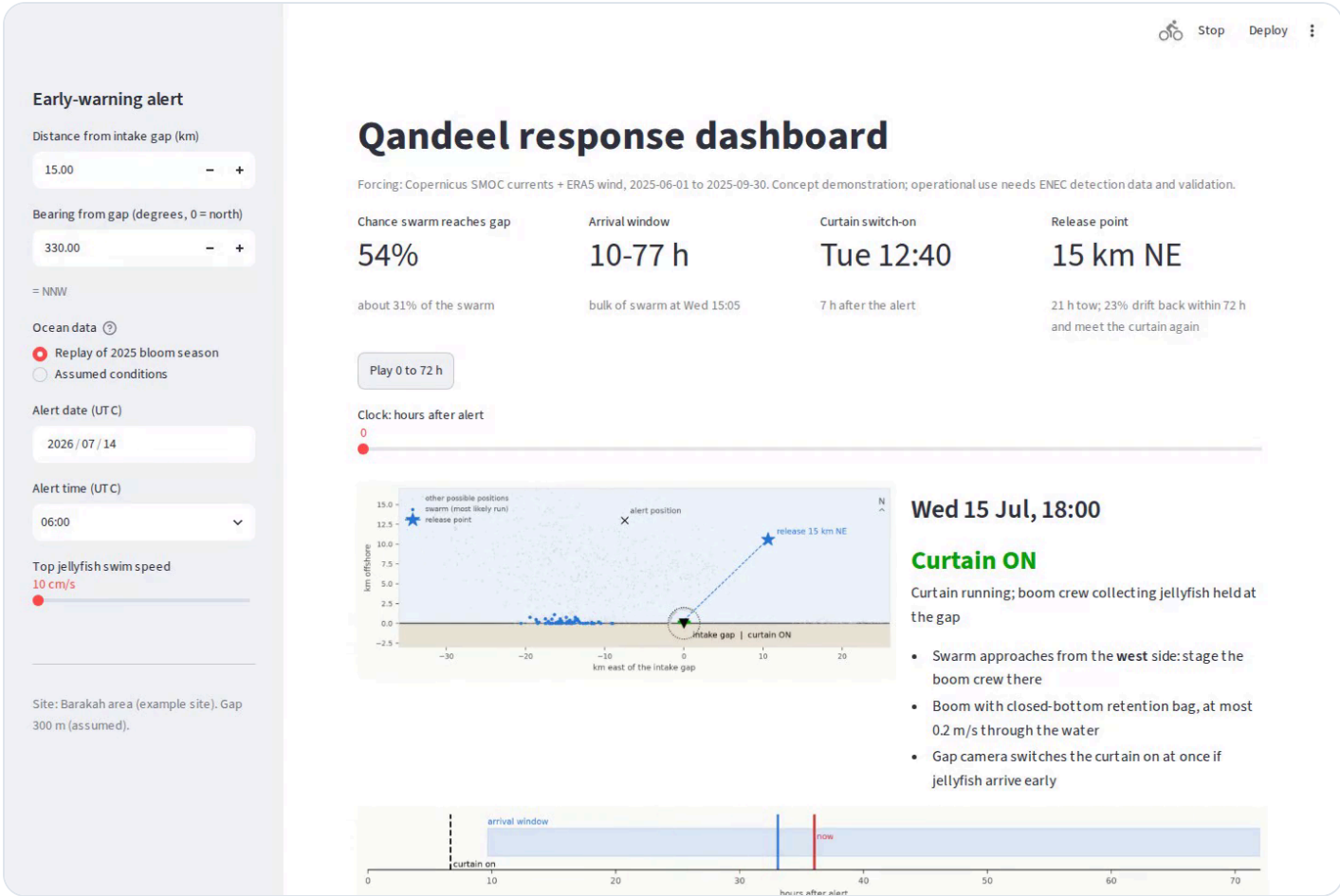
Two boats close a U-boom with a closed-bottom retention bag around them.
- 6

Release.

Move with the current at  $\leq 0.2$  m/s through the water; open the boom down-current.
- 7

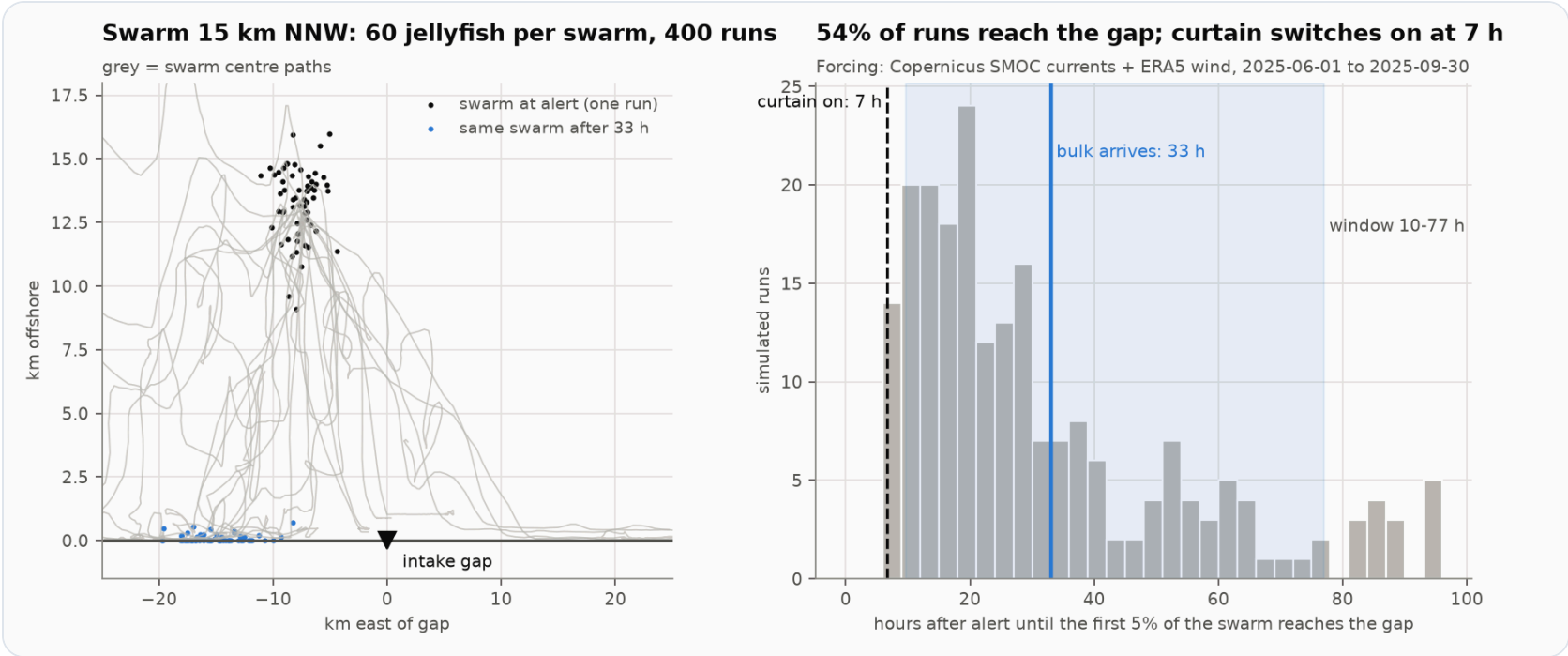
Learn.

Camera and sensor data improve the next plan. Backup: the gap camera starts the curtain at once if jellyfish arrive early.

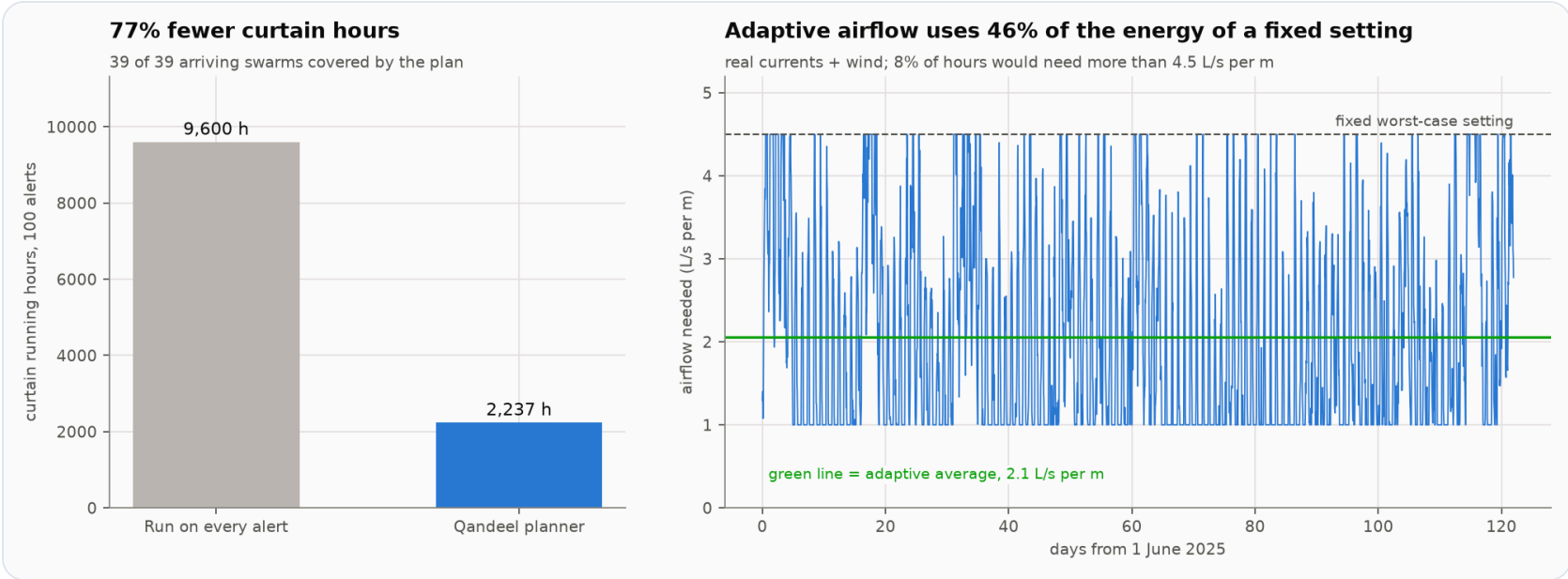


Qandeel operator dashboard (working prototype, Streamlit): chance of arrival, window, switch-on clock time, swarm map, curtain status, release point. Data: Copernicus SMOC currents + ERA5 wind, summer 2025.

# The planner runs the curtain only when it matters



Swarm 15 km NNW: 400 runs on real June–September 2025 conditions.



Hindcast of 100 alerts at random times and places in the real 2025 season.

## 39 / 39 covered

Every swarm that truly reached the gap was covered by the plan; the curtain stayed off for 45 of 61 that did not arrive.

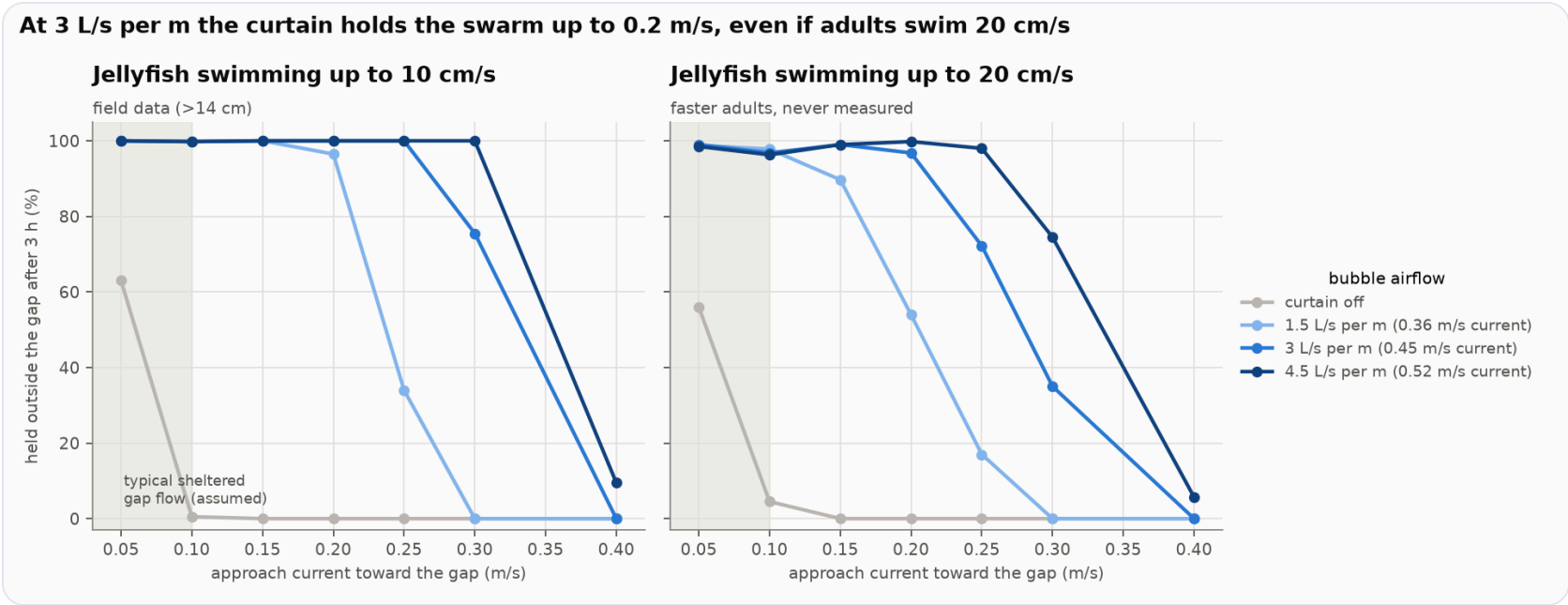
## 77% fewer hours

Versus switching on for every alert. A live forecast narrows the window further (10–77 h → ~10–20 h in tests).

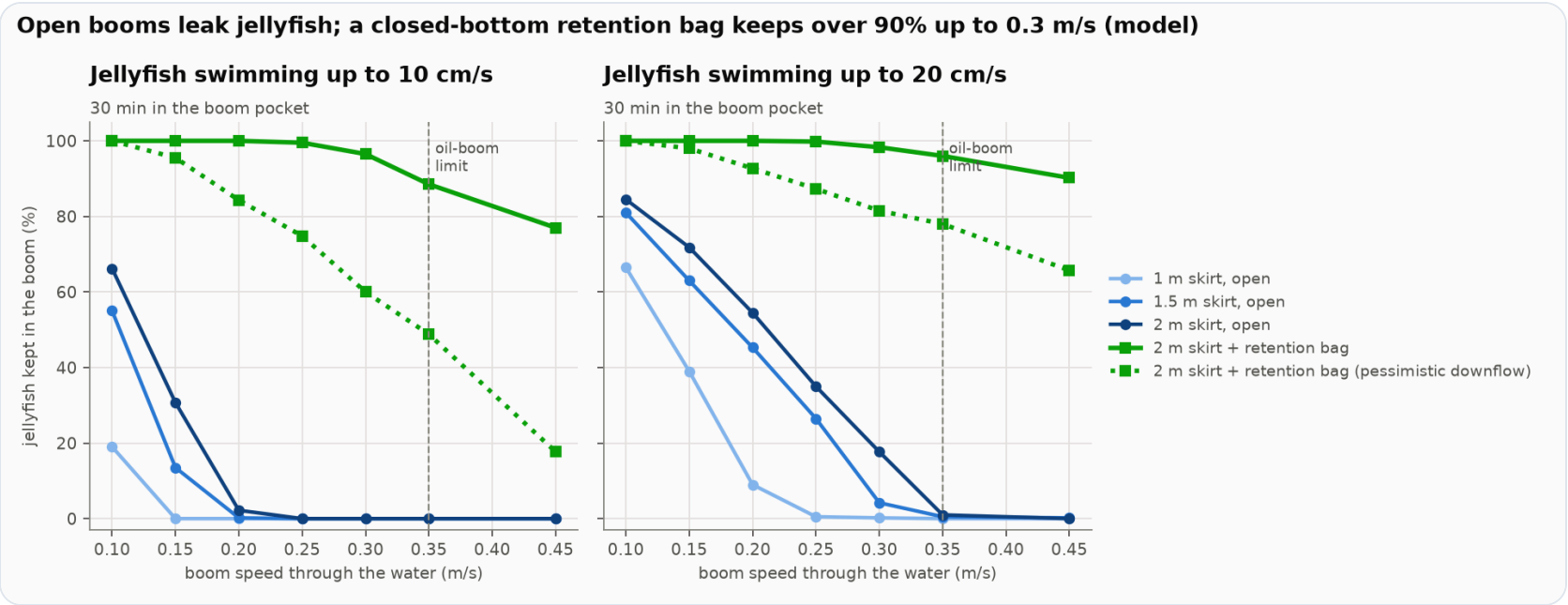
## 46% of the energy

Adaptive airflow follows real currents and wind. ~8% of hours need more air than the design maximum: flagged for the crew.

# Our finding: oil-spill booms leak jellyfish, and a bag fixes it



400 individually simulated jellyfish per point; surface current from Bulson's pneumatic-barrier relation.



Jellyfish depth from field data; apex downflow 25% (pessimistic 50%) of tow speed.

## Curtain holds the swarm

At 3 L/s of air per metre it holds 100% up to 0.2 m/s approach current, 95% even if adults swim 20 cm/s; real currents average 0.08 m/s. It fails above ~0.3 m/s, matching field trials.

## Why the bag matters

Oil floats back up; neutrally buoyant jellyfish are dragged under the skirt far below the 0.35 m/s oil-boom limit (0–54% kept at 0.2 m/s). A closed-bottom fabric bag keeps 84–100% in the model.

# Gentle on marine life, built from parts that exist today

## Marine-life protection

| RISK                          | HOW WE REDUCE IT                                                                         |
|-------------------------------|------------------------------------------------------------------------------------------|
| Jellyfish killed              | Kept in the water and released alive; no cutting, pumping or mesh                        |
| Turtles, dolphins in the boom | Escape gap, slow tow, camera watch, stop on sighting                                     |
| Seabed, sediment              | Pipe on legs 0.5–1 m above the bed, routed away from coral and seagrass                  |
| Bubble noise, fish            | Runs only during events; measured in the pilot                                           |
| Released swarm returns        | Release point simulated for 72 h; returners meet the curtain again (~1.3× load)          |
| Carbon                        | ~11 MWh per event: ~4.4 t CO <sub>2</sub> on a gas grid, ~0.1 t on the plant’s own power |

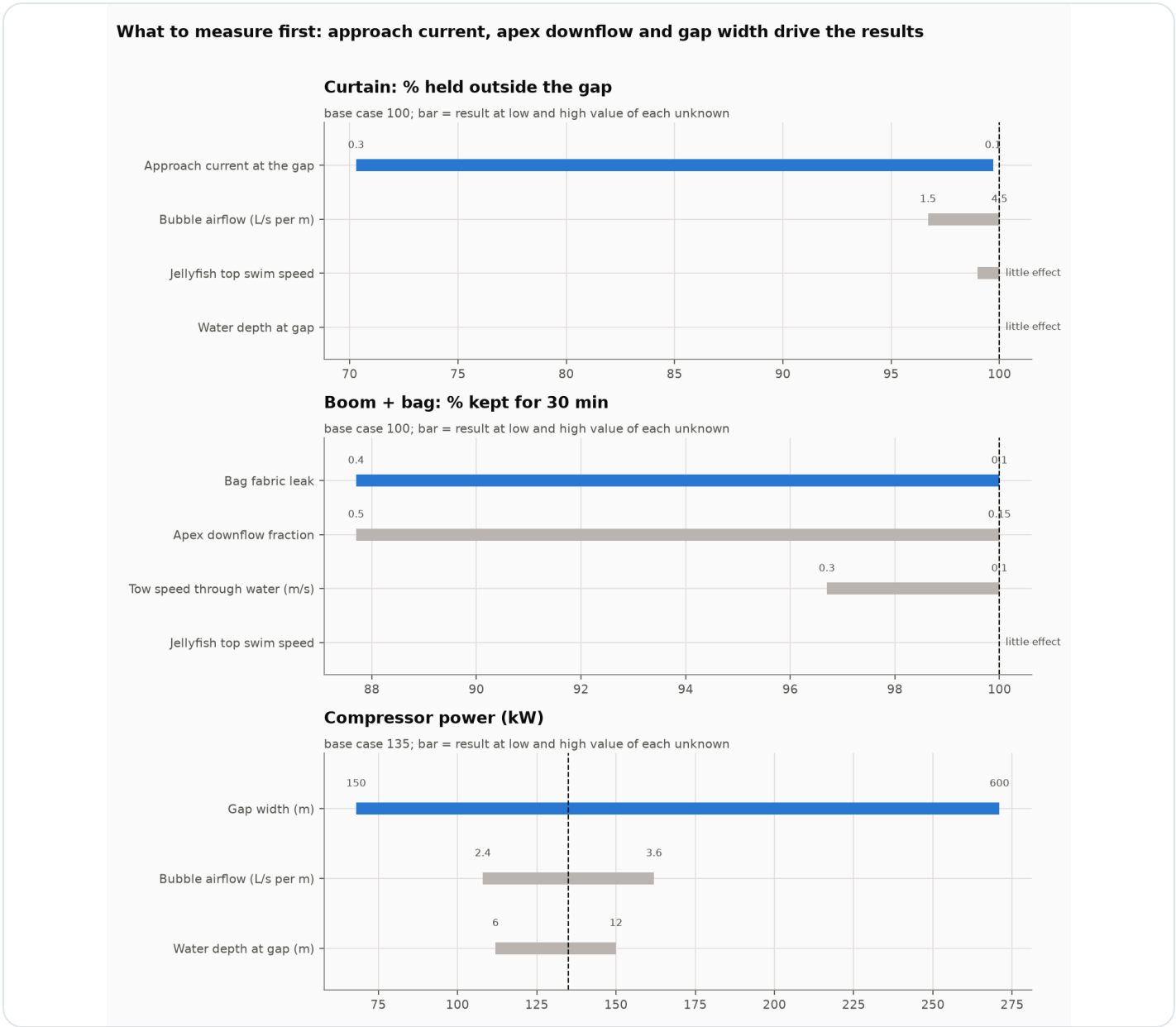
## Feasibility

| ITEM                 | ESTIMATE (300 M GAP)                                                                  |
|----------------------|---------------------------------------------------------------------------------------|
| Compressed air       | 0.9 m³/s, ~135 kW, oil-free (ISO 8573-1 Class 0)                                      |
| Capital (indicative) | USD 0.4–1.3 M: curtain, compressor + standby, boom, bags, boats, sensors              |
| Running cost         | ~USD 1,100 of energy per event                                                        |
| Value                | One avoided 12 h unit outage (~USD 0.5 M, EPRI figure) pays back a third to all of it |
| Capacity             | One boom pair: ~2,200–21,600 jellyfish/h (4–43 t/h) vs 0.9 t/h for a cutting robot    |
| Gulf hardening       | Membrane diffusers, foul-release coating, modular sections, lifted out off-season     |

# What we assume, what we don't know yet, and what we measure first

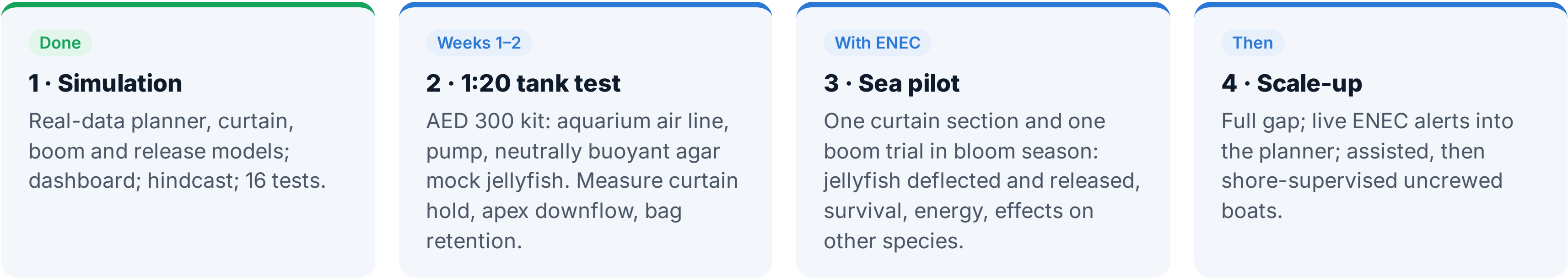
| ASSUMPTION                                                   | BASIS                             | IF WRONG                                         |
|--------------------------------------------------------------|-----------------------------------|--------------------------------------------------|
| Intake behind a breakwater with one gap, ~300 m, ~9.5 m deep | Brief; public channel depth       | Curtain size scales; measure on satellite images |
| Alert gives position, hours of notice                        | Brief: "time to prepare"          | Camera trigger switches the curtain on           |
| Swarm in top 1 m; adults swim $\leq 10\text{--}20$ cm/s      | Field and lab studies (Australia) | Deeper skirt; tested at 20 cm/s                  |
| Swarm density 0.1–1 per m <sup>3</sup>                       | No published figure               | Capacity scales linearly                         |
| Apex downflow 25–50% of tow speed; bag passes 20%            | Model hypothesis                  | Measured in the tank test                        |

**Limitations:** no management method has been tested on Blue Blubbers; bubble curtains weaken in strong currents (~8% of hours need extra air); ~23% of released jellyfish may drift back within 3 days; uncrewed boats are a later phase.



Sensitivity: approach current, bag leakage / apex downflow and gap width drive the results, so we measure them first.

# From simulation to sea, one measured step at a time

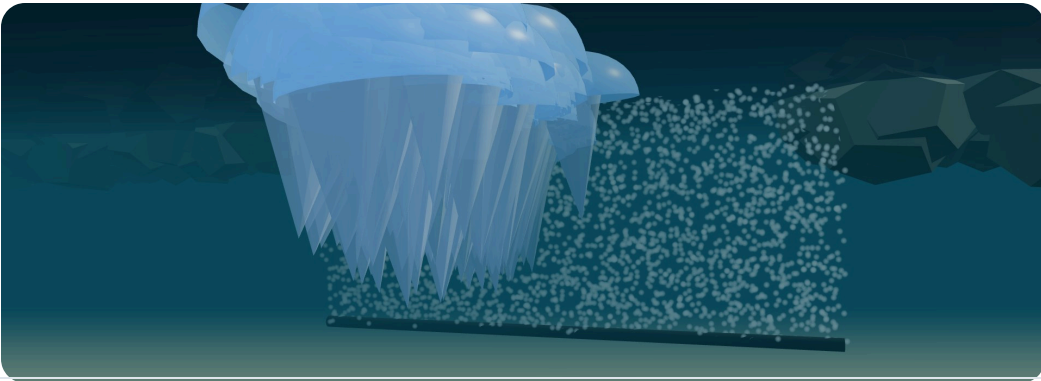
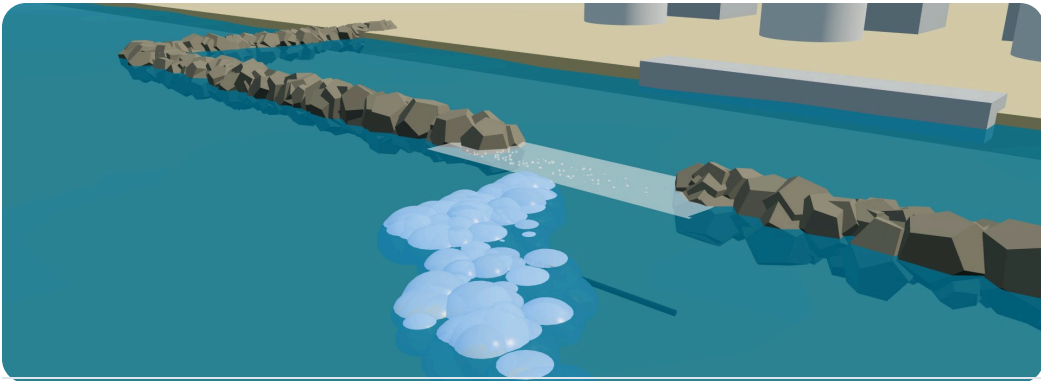


## Pass marks

- Tank:  $\geq 95\%$  held at the scaled 0.2 m/s current
- Tank: bag keeps  $\geq 90\%$  at the scaled 0.2 m/s tow
- Pilot: no harm to non-target species; fewer screen-clearing hours

## Open work and AI use

Code, data and tests: [github.com/Shaaha-7/ENEC](https://github.com/Shaaha-7/ENEC). AI assistants helped search and summarise literature, check claims, draft text and write simulation code, and an open text-to-speech model voices the video; the team reviewed every decision, assumption and figure, and can explain each in our own words.



Stills from our 3D explainer video: the curtain holding a swarm outside the gap, and the bubble wall seen underwater.